Epistemic Situational Awareness (ESA)

Continuation Governance for Frontier AI Systems

When ESA is present, inadmissible moves are not merely filtered or refused - they cease to arise as candidates.

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Abstract

Contemporary reasoning systems - human, institutional, and machine - exhibit increasing generative power without a corresponding ability to locate themselves epistemically. As a result, activity persists after admissibility has collapsed: provisional repairs harden into explanations, explanations into memory, and memory into authority without warrant.

This paper introduces **Epistemic Situational Awareness (ESA)** as a system-level control property that governs *when reasoning may continue, escalate, bind, or lawfully terminate.* ESA is not a cognitive faculty, belief model, or form of consciousness. It is an emergent orientation produced when epistemic governance is enforced strongly enough to shape which continuations even appear possible.

When ESA is present, inadmissible moves are not merely filtered or refused - they cease to arise as candidates. The paper defines ESA, distinguishes it from adjacent concepts, characterizes its behavioral signatures and failure modes, and explains what ESA enables that current reasoning systems lack. ESA does not add intelligence or optimize outcomes. It adds **jurisdiction**: the ability to know *what kind of epistemic act is occurring, what authority it would create if reused, and when stopping is the most disciplined move available.*

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1. Introduction

Failures in modern reasoning systems rarely stem from insufficient intelligence. They stem from **unlocated continuation**: reasoning proceeds without a stable sense of epistemic role, admissible evidence, promotable status, or lawful endpoints.

This pathology appears across domains: large language models that continue producing plausible narratives after evidentiary exhaustion; institutions where

provisional practices silently become binding doctrine; and human–AI workflows where successful execution substitutes for understanding.

Most mitigations address outputs - filters, guardrails, refusals, confidence labeling - without addressing whether *continuation itself is warranted*. This paper governs something prior: the conditions under which reasoning is permitted to move, escalate, bind, or stop.

The framework governs **movement of epistemic artifacts**: who may move what kind of artifact, from which role to which role, under what warrants, and with what consequences if reused.

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2. Definition: Epistemic Situational Awareness

Epistemic Situational Awareness (ESA) is the continuous ability of a reasoning system to recognize:

1. what kind of epistemic act it is performing
2. which epistemic layer the artifact it is acting on occupies
3. what authority that artifact would acquire if reused
4. what escalations of epistemic privilege are currently admissible
5. when lawful termination - including non-decision - has been reached

ESA is not knowledge of the world. It is knowledge of **epistemic position**. A system with ESA reasons with awareness of *what its reasoning is allowed to become*.

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3. What ESA Is Not

ESA is not introspection, belief modeling, confidence estimation, self-reflection, or consciousness. It does not require access to internal representations or subjective experience. The term *situational awareness* is used in its engineering sense, as in aviation and safety-critical systems: awareness as **constraint on action**, not awareness of meaning.

ESA therefore denotes a **control property**, not a mental state.

Governance under ESA applies only to **epistemic artifacts** - texts, traces, decisions, memories, instructions, or policies eligible for reuse. Internal reasoning and latent representations are unconstrained until externalized into such artifacts.

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4. ESA as an Emergent Control Property

ESA is not installed directly. It **emerges** when epistemic governance is enforced consistently enough to alter a system's action horizon. In practice, this requires:

- explicit separation between epistemic **modes** of reasoning
- typed epistemic actions with bounded scope
- declared observation policy
- fail-closed promotion semantics
- lawful termination states, including *Blocked* and *Undecidable*

4.1 Epistemic Modes

Epistemic modes classify *the intent of an epistemic act*, not its correctness, confidence, or layer. The standing modes are:

- **Exploration** - generating candidate ideas or hypotheses without authority to bind or explain
- **Refinement** - narrowing or improving candidate structures without asserting correctness
- **Repair** - modifying an existing artifact to restore known behavior without explaining cause
- **Expansion** - promoting an artifact into a role with greater downstream reuse or authority

Modes do not imply movement across epistemic layers. They describe *how* an artifact is being worked on, not *what kind of thing* the artifact is.

4.2 Epistemic Layers

Epistemic layers describe the **ontological role** an artifact occupies:

- **Specification** - what counts as correctness or meaning independent of execution
- **Configuration** - which degrees of freedom are currently fixed to produce behavior
- **Implementation** - concrete execution, artifacts, and observable behavior

Layer transitions are separate, governed acts. No mode, by itself, authorizes movement between layers.

When these constraints are applied reliably, a qualitative shift occurs: **inadmissible moves do not merely become forbidden; they cease to appear as candidates**. This distinguishes ESA from post-hoc filtering or refusal systems, which still generate inadmissible continuations and suppress them later.

Operationally, ESA is enforced at the interface between generation and system-level acts - memory writes, escalations, bindings, or external actions - rather than within token selection itself.

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5. Behavioral Signatures

5.1 Lawful Termination

In systems without ESA, uncertainty triggers further generation. Activity substitutes for progress. With ESA, termination is a correct epistemic outcome. Silence is not absence; it is a formally valid state under fixed observation policy.

5.2 Separation of Execution and Understanding

Execution success is a fact; understanding is a claim. ESA enforces this distinction operationally. Restored behavior does not self-promote into explanation without explicit authorization.

5.3 Situated Refusal

Refusal under ESA is not brittle. It is situated: a structured declaration that a move is inadmissible *here*, under current mode and layer, coupled with a legible path to escalation if warranted.

5.4 Reduced Thrash

Most agent thrashing is epistemic, not computational. ESA introduces epistemic exhaustion as a visible state, reducing cycles that add no new warrant.

5.5 Oversight with Leverage

ESA creates clear procedural hooks for oversight - authorize escalation, contest promotion, modify observation policy, accept or reject termination - without inspecting raw reasoning.

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6. Responsibility as Admissibility

ESA reframes responsibility as **jurisdictional boundary crossing under declared constraints**. Accountability attaches not to intent or outcome, but to escalation, binding, or reuse without admissible warrant. This localization moves responsibility upstream of harm and scales across humans, institutions, and machines.

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7. Authority Laundering and Time Decay

A primary harm vector is **authority laundering**: exploratory or provisional artifacts accrue authority through reuse over time. The artifact remains unchanged; its role changes.

In this paper, **binding** refers to the procedural act by which an artifact is permitted to acquire ongoing epistemic authority - such as reuse without re-derivation, reference as precedent, enforcement, or institutional memory. Binding is not belief, agreement, or semantic truth. It determines what consequences reuse may legitimately carry.

ESA blocks illegitimate binding by requiring explicit authorization at each jurisdictional boundary.

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8. Failure Modes

ESA can degrade in predictable ways: ritualized enforcement, over-pruning, shadow escalation through untyped channels, and mode confusion. These failures function as diagnostics rather than refutations.

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9. A Unifying Grammar

ESA can be operationalized through four standing questions:

1. **Mode** - What kind of epistemic act is occurring?
2. **Evidence** - What observation is admissible here?
3. **Binding** - What authority would reuse create?
4. **Termination** - How does this end lawfully?

These questions do not guarantee correctness. They guarantee **orientation**.

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10. Conclusion

Epistemic Situational Awareness names a missing capability in modern reasoning systems: the ability to locate oneself epistemically and act accordingly. ESA does not ask systems to be wiser. It teaches them **jurisdiction**.

Correctness, under ESA, is not the right answer. It is the **right to answer at all**.

Corpus-Wide Reference Positioning

This paper argues that contemporary reasoning systems - human, institutional, and machine - exhibit increasing generative power without a corresponding ability to locate themselves epistemically. Its nearest external comparison fields are consciousness, governance, ai. The sources in [E1], [E2], [E3], [E4], [E5],

[E6], [E7] are cited as predecessor literature, adjacent standards, or counter-pressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Structural and computational correspondences remain insufficient as phenomenal evidence unless linked to discriminating empirical predictions against competing theories. A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. Retrieval, tool use, and policy layers can reduce unsupported output without eliminating model error, source error, omission, bias, or the environmental and institutional costs of scale. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[E1] David J. Chalmers. "Facing Up to the Problem of Consciousness." *Journal of Consciousness Studies* 2(3), 1995, pp. 200-219. https://consc.net/papers/facing.html
Relationship: Foundational framing of the explanatory gap; cited to bound, not resolve, phenomenal claims.

[E2] Larissa Albantakis et al. "Integrated Information Theory (IIT) 4.0: Formulating the Properties of Phenomenal Existence in Physical Terms." *PLOS Com-*

putational Biology 19(10), 2023, e1011465. DOI: 10.1371/journal.pcbi.1011465. https://doi.org/10.1371/journal.pcbi.1011465 **Relationship:** Competing formal theory of consciousness included for explicit contrast and counterevidence obligations.

[**E3**] Stanislas Dehaene and Jean-Pierre Changeux. "Experimental and Theoretical Approaches to Conscious Processing." *Neuron* 70(2), 2011, pp. 200-227. DOI: 10.1016/j.neuron.2011.03.018. https://doi.org/10.1016/j.neuron.2011.03.018 **Relationship:** Competing global-workspace account included to prevent a corpus-native ontology from substituting for empirical neuroscience.

[**E4**] Nikolaus Kriegeskorte, Marieke Mur, and Peter A. Bandettini. "Representational Similarity Analysis - Connecting the Branches of Systems Neuroscience." *Frontiers in Systems Neuroscience* 2, 2008, 4. DOI: 10.3389/neuro.06.004.2008. https://doi.org/10.3389/neuro.06.004.2008 **Relationship:** Predecessor empirical method for comparing representational geometries.

[**E5**] Francesco Locatello et al. "Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations." *PMLR* 97, 2019, pp. 4114-4124. https://proceedings.mlr.press/v97/locatello19a.html **Relationship:** Counterevidence showing that desired latent factorization is not identifiable from unsupervised data without inductive biases.

[**E6**] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action.* Cambridge University Press, 1990. DOI: 10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763 **Relationship:** Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E7**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/ **Relationship:** Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

Ableman corpus

[**A1**] Adam Ableman. *Ableman Consciousness Ontology (ACO).* Ableman Corpus current edition; work identity `PAPER.PUB.18203749`;
source SHA-256
`ACD97032780DA7B3A36750702BE1F47BCB2E6361EF6E74ECA8B441D7C6A25DDE`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *The Locus of Qualia.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-AF27E66B81811DB1`;
source SHA-256
`7B83EBD0FFDCFEBFA522A715AE0E4D1049FCA0B4CE0618C2017F8BB9206E5424`.

Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *The Phenomenal Aperture.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-4B275CBB7EFF1A4F`;
source SHA-256
`B6AE95D72B89DAF47778BED3982843A1ECE350CA363750108AA60D5A4C63D763`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Possible Minds: Against the Provincialization of Consciousness.* Ableman Corpus current edition; work identity `PAPER.PUB.20681869`;
source SHA-256
`77F381A29E2EF801085938DB369F3F432321CCEB8B9DD3969330337C24BA4998`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Admissibility Theory of Consciousness.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-8FDC994841DA6CF3`;
source SHA-256
`F409652555CEC4898DBEC1E8BBB179CEE020E7FF35F4C8F307CDA82253D6F49F`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Ableman Constitutional Governance.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-79353AF163A4B0F1`;
source SHA-256
`FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Constitutional Theory.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D2D97C4042FD281F`;
source SHA-256
`FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Editorial source notes retained from predecessor

1. Ableman Adam Ableman *Governed Coding and Debugging: A Claim-Governance Grammar for Human–AI Agentic Workflows.* 2025.

2. Ableman Adam Ableman *Ableman's Razor: Epistemic Authority and Illegitimate Claim Migration*. 2024.
3. Ableman Adam Ableman *Constitutional Epistemics (CE): Governance of Epistemic Authority in Hybrid Systems*. 2024.
4. Endsley, M. R. "Toward a Theory of Situation Awareness in Dynamic Systems." *Human Factors*, 1995.
5. Floridi, L., et al. "AI Governance and the Problem of Control." *Philosophy & Technology*, 2018.
6. Amodei, D., et al. "Concrete Problems in AI Safety." arXiv:1606.06565.
7. Klein, G. "A Recognition-Primed Decision Model of Rapid Decision Making." *Military Operations Research*, 1993.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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